SportsDeck: Database Model Design

By Haifa Khan and Raweeha Fatima Raza

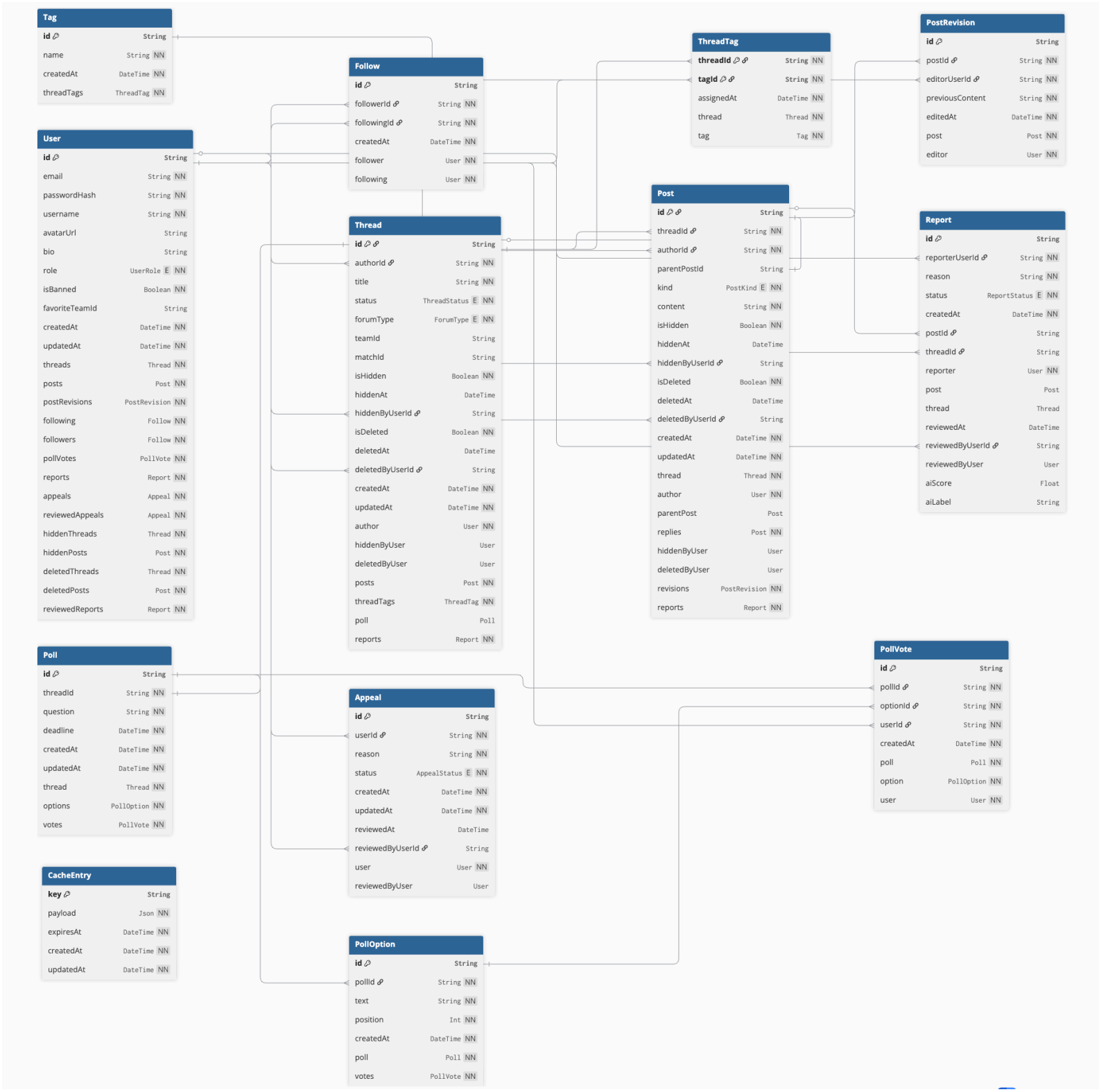


Figure 1: Entity-Relationship Diagram of the SportsDeck database schema

User: Represents a registered account. Stores login credentials (hashed password), profile info (username, avatar, bio), role (REGULAR or ADMIN), ban status, and an optional favourite team. A user can author threads and posts, follow other users, vote in polls, submit reports, and file ban appeals.

Follow: Represents a directional follow relationship between two users. Stores the follower and the user being followed, along with the timestamp of the follow action. Enforces uniqueness so a user cannot follow the same person twice.

Thread: Represents a discussion thread in either a general forum or a team-specific forum. Stores the title, forum type, optional team, and match associations, open/closed status, and soft-delete/hidden flags. A thread can have posts, tags, a poll, and reports.

Tag: Represents a label that can be applied to threads for categorization and search. Tag names are unique across the system.

ThreadTag: Join table linking threads to tags. Represents a many-to-many relationship between Thread and Tag.

Post: Represents a message within a thread. Can be a root post (the opening message) or a reply. Supports nested replies via a self-referential parent relationship. Stores content, soft-delete/hidden flags, and tracks edit history through revisions.

PostRevision: Stores previous versions of a post whenever it is edited. Allows users to view and edit history. Each revision records the previous content, who edited it, and when.

Poll: Represents a poll attached to a thread. Has a question, a deadline after which voting is disabled, and multiple options. Each thread can have at most one poll.

PollOption: Represents a single votable option within a poll. Has a display position for ordering.

PollVote: Records a user's vote on a poll option. Enforces one vote per user per poll.

Report: Represents a user-submitted report flagging a post or thread as inappropriate. Stores the reason, current status (PENDING, DISMISSED, APPROVED), and optional AI-generated toxicity score and label used to assist admin review.

Appeal: Represents a ban appeal submitted by a banned user. Stores the reason, status (PENDING, APPROVED, REJECTED), and which admin reviewed it.

CacheEntry: A generic key-value cache table used to store responses from external APIs (sports data, AI inference) to avoid redundant requests and stay within free-tier limits. Each entry has an expiry timestamp.